ARDUINO UNO PROJECT

Automated Smart Railway Crossing and Obstacle Detection System

Ultrasonic sensor, servo gate, LEDs, buzzer and LCD

Presented by

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Why this project?

Problem

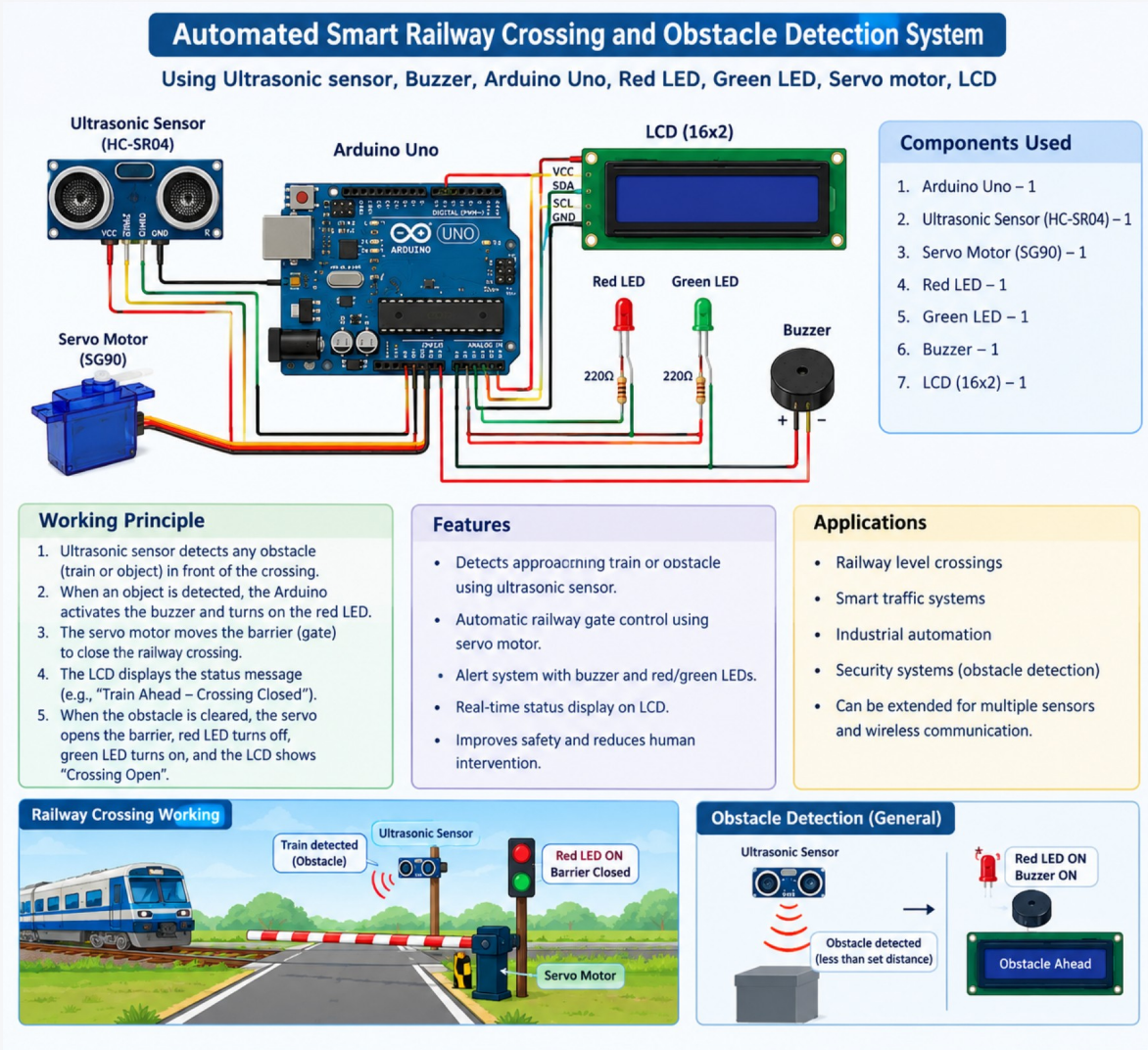
Many level crossings depend on a person to close the gate. A delay or a mistake can cause accidents.

Our solution

An Arduino senses the train with an ultrasonic sensor and closes the gate, warns with light and sound, and shows a message.

Project overview

- Sensor detects the train
- Arduino decides
- Gate, LEDs, buzzer and LCD respond



Components used

Component	Job in the project
Arduino Uno	The brain: reads sensor, controls everything
Ultrasonic sensor HC-SR04	Measures distance to train or obstacle
Servo motor	Opens and closes the gate
Red LED	Danger: train is coming
Green LED	Safe: track is clear
Buzzer	Warning sound
LCD 16x2	Shows the status message

How it works

Train detected

Distance is less than the set limit
Red LED ON
Buzzer ON
Servo closes the gate
LCD: Train Ahead

Track clear

Distance is more than the set limit
Green LED ON
Buzzer OFF
Servo opens the gate
LCD: Track Clear, Gate Open

The Arduino checks the distance again and again, so it always updates.

My Proteus simulation

1. Circuit

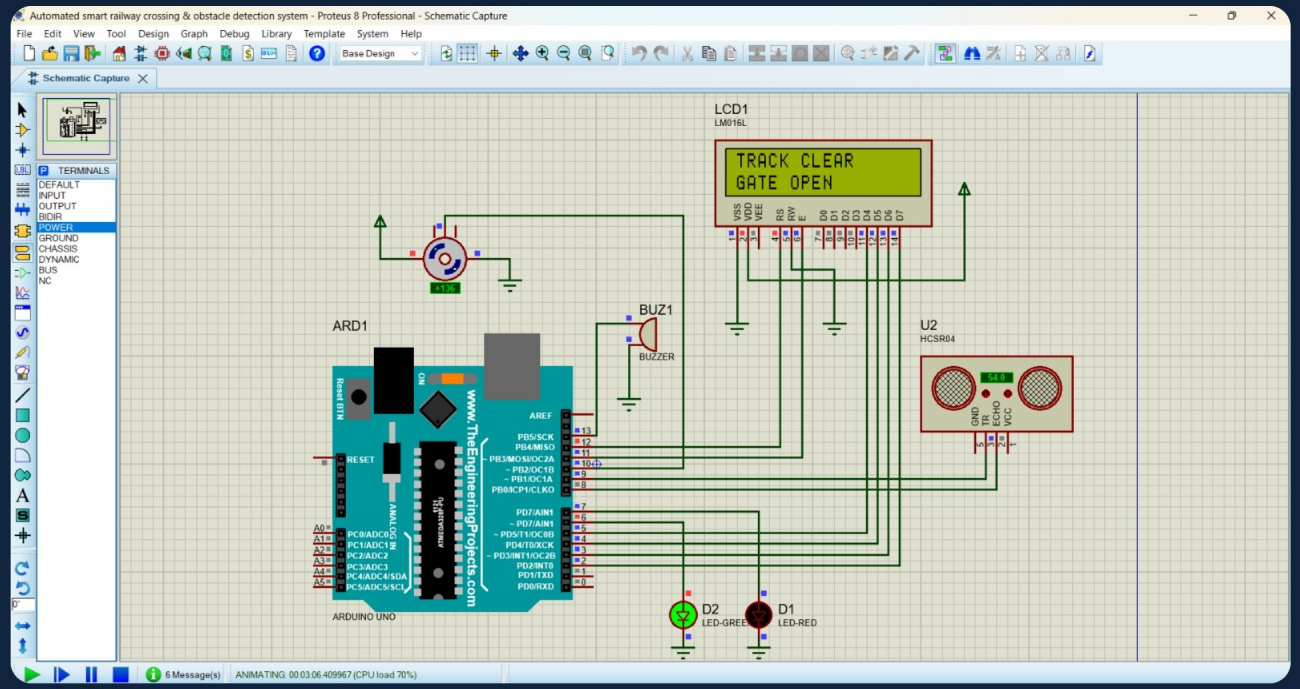
Arduino with sensor, LCD, servo, buzzer and 2 LEDs.

2. Input

Sensor shows 54 cm, so no train is near.

3. Output

LCD: TRACK CLEAR, GATE OPEN. Green LED ON, red LED OFF.



Features and applications

Features

- Fully automatic gate
- Light and sound alert
- Live status on LCD
- Low cost, less human work

Applications

- Railway level crossings
- Smart traffic systems
- Industrial automation
- Security obstacle detection

Conclusion

A low-cost Arduino system that detects a train, closes the gate and warns people, making level crossings safer.

Thank you. Any questions?